

FlagAI: A Referee Game-Day Assistant

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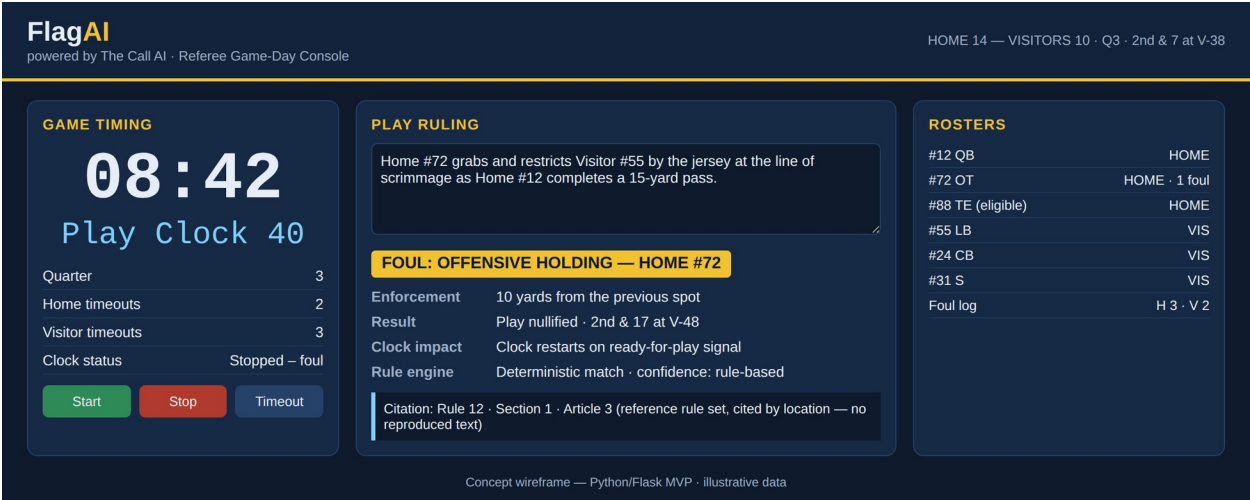


Figure 1. Concept wireframe of the FlagAI referee game-day console showing game timing, play ruling with rule citation, and rosters (illustrative data).

Technical Field

This project involves the technical fields of Artificial Intelligence (specifically rule-based expert systems and knowledge representation), Web Application Development, and Sports Technology, with a focus on decision support for football officiating.

Background Information

Football officials must identify a foul, determine its enforcement, and manage the game clock within seconds, while applying a professional rulebook organized into 19 rules, each divided into sections and articles (National Football League, 2026). Officiating errors at the professional level draw national scrutiny, and at the amateur level, crews frequently work without replay support or a rules expert on site. The inspiration for this project came from the creator’s prior coursework research on artificial intelligence in professional football officiating, combined with 13 years of service as a U.S. Navy Electrician’s Mate, where standardized technical manuals and checklists guide time-critical decisions. That experience established the core premise of the project: officiating benefits from a fast, authoritative reference that returns a traceable answer rather than an opinion. The concept began as a multi-agent penalty-call assistant. Peer feedback and subject-matter-expert review then refocused it into a referee-facing game-day application that combines

rule determination with roster and game-clock management, because rule lookups alone are already available through general-purpose AI tools.

Prior Art (Research)

Prior Art Statement: This project builds upon existing published rule references, officiating software, and automated officiating technologies. The works below represent the prior art related to this innovation. Each entry describes the existing product or study and explains how FlagAI differs from it.

2026 NFL Rulebook (NFL Football Operations)

Description: The official playing rules for professional football are published online and as a downloadable PDF, organized by rule, section, and article, with penalty summary tables and field diagrams. The rulebook is the authoritative reference source that officials study and consult.

Differentiation: The rulebook is a static document that requires the official to locate and interpret the correct provision. FlagAI encodes rule logic as original, deterministic code that accepts a play description and returns the foul, enforcement, and a rule-section-article citation without reproducing the rulebook text.

National Football League. (2026). *2026 NFL rulebook*. NFL Football Operations.
<https://operations.nfl.com/the-rules/nfl-rulebook/>

NFHS Digital

Description: NFHS Digital is the National Federation of State High School Associations' mobile platform that gives officials, coaches, and administrators access to official rules books and case books for 18 high school sports, with search, bookmarks, highlighting, and notes.

Differentiation: NFHS Digital is a reading and search tool; the official must still find and interpret the rule. FlagAI produces a ruling, calculates enforcement, and connects that ruling to the game clock and team rosters.

National Federation of State High School Associations. (2025). *NFHS Digital* [Mobile app]. App Store. <https://apps.apple.com/us/app/nfhs-digital/id6746011559>

REFSIX

Description: REFSIX is a soccer referee application for smartphones and smartwatches that provides multiple match timers, logs cautions and dismissals, tracks substitutions and rosters, generates match reports, and records referee performance analytics.

Differentiation: REFSIX is the closest analog to FlagAI's timing and roster features, but it serves soccer only and offers no rule interpretation. FlagAI brings game-day timing and roster tracking to football and links them to a rule engine, so each ruling updates the foul log and clock status automatically.

REFSIX. (n.d.). *REFSIX: The app for football/soccer referees*. <https://refsix.com/>

Arbiter Referee Assigning Software

Description: Arbiter is an officials-management platform used by more than 10,000 organizations and schools to assign officials to games, track availability, and process payments electronically.

Differentiation: Arbiter manages officiating logistics before and after a game but provides no support during play. FlagAI operates during the game, supporting rulings, rosters, and timing in real time.

Arbiter. (n.d.). *Referee assigning software*. <https://arbiter.io/referee-assigning-software/>

Hawk-Eye Line-to-Gain Measurement

Description: Beginning with the 2025 season, the NFL adopted Sony's Hawk-Eye optical tracking system, which uses six 8K cameras to determine whether the ball has reached the line to gain, with measurements coordinated through the league's central officiating center.

Differentiation: Hawk-Eye requires stadium camera infrastructure and addresses a single measurement. FlagAI is software-only, runs on a standard computer or tablet, and addresses foul determination and enforcement across the rule set.

National Football League. (2025). *Sony's Hawk-Eye Innovations selected by NFL to revolutionize line to gain measurements*. NFL.com. <https://www.nfl.com/news/sony-s-hawk-eye-innovations-selected-by-nfl-to-revolutionize-line-to-gain-measurements>

FIFA Semi-Automated Offside Technology

Description: Semi-automated offside technology uses 12 tracking cameras that follow up to 29 data points per player at 50 frames per second, and artificial intelligence combines limb and ball data to alert video match officials to potential offside positions.

Differentiation: The system is vision-based, limited to one rule, and available only in elite venues. FlagAI is text-based, covers many rules, and is accessible to officials at any level without specialized hardware.

FIFA. (2022). *Semi-automated offside technology*. Inside FIFA. <https://inside.fifa.com/innovation/world-cup-2022/semi-automated-offside-technology>

ChatGPT and General-Purpose Large Language Models

Description: Conversational AI models answer rules questions in natural language and can summarize or explain football rules on request.

Differentiation: Language model output is probabilistic; the same question can produce different answers, and cited rule numbers are not guaranteed to be accurate. FlagAI's rulings come from a deterministic rule engine that returns the same verifiable result for the same input, with any language model role limited to advisory support; general chatbots also do not manage game clocks or rosters.

OpenAI. (2026). *ChatGPT* [Large language model]. <https://chatgpt.com/>

Possibilities of Artificial Intelligence in Sports Refereeing (Research Study)

Description: This exploratory study compares the published literature on AI in sports refereeing with opportunities identified by officiating experts, finding that research concentrates heavily on accuracy systems while other areas, such as athlete safety, remain largely unexplored.

Differentiation: The study documents a gap between AI's potential in officiating and its practical implementation. FlagAI addresses part of that gap by providing accessible decision support to officials rather than camera-based accuracy tools.

Martín Moncunill, D., Sampedro Lirio, D., & Bravo Hijón, M. Á. (2026). Possibilities of artificial intelligence in sports refereeing: An exploratory study contrasting the literature review with expert-perceived opportunities. *Multimodal Technologies and Interaction*, 10(3), 30. <https://doi.org/10.3390/mti10030030>

AI Is Helping Referee Games in Major Sports Leagues (Scientific American)

Description: This article reviews AI officiating tools in tennis, baseball, and basketball and describes their limitations, including speed-accuracy tradeoffs and visual interference, concluding that experts view the technology as an aid to human decision-making rather than a replacement.

Differentiation: The article reinforces FlagAI's design principle: the official retains final authority, and the application supplies a fast, citable recommendation that supports the official's judgment.

Dowsett, B. (2024, May 2). AI is helping referee games in major sports leagues, but limitations remain. *Scientific American*. <https://www.scientificamerican.com/article/ai-is-helping-referee-games-in-major-sports-leagues-but-limitations-remain/>

Project Description

FlagAI, powered by The Call AI, is a web-based game-day assistant for football officials, built in Python with the Flask framework. The application operates in the area of sports officiating decision support and combines three functions in a single console. First, a deterministic rule engine accepts a plain-text description of a play and returns the applicable foul, the responsible player, the enforcement (yardage, enforcement spot, and resulting down and distance), and a citation by rule, section, and article; the rule logic is written as original code rather than reproduced rulebook text. Second, a roster module stores team rosters so that each foul is attributed to a specific player number and position and recorded in a foul log. Third, a game-timing module manages the game clock, play clock, quarters, and timeouts and reflects the clock consequences of each ruling. The intended user population includes on-field game officials and officiating crews, officiating trainees and clinic instructors, and league or association rules coordinators. The minimum viable product uses professional football playing rules as its reference rule set, and the rule engine is modular so that additional rule sets, such as high school rules, can be added.

Innovation Claim

This project is innovative because it unites a deterministic, citation-producing football penalty rule engine with roster and game-clock management in a single referee-facing application, so every ruling is repeatable, traceable to a specific rule, section, and article, and immediately reflected in the game state. Unlike general-purpose AI chatbots or static digital rulebooks, FlagAI returns the same verifiable answer for the same play every time.

Usage Scenario

Beyond live game-day use, FlagAI serves as an officiating training and evaluation platform. An officiating association conducting a preseason clinic loads a set of written play descriptions drawn from game film, trainees record their own rulings, and each ruling is compared against the rule engine's determination and citation. Because the engine is deterministic, every trainee is measured against the same standard, and instructors can identify which rule areas, such as pass interference or enforcement spots, produce the most errors across the group. Officiating crews can apply the same workflow to post-game review by entering disputed plays to confirm the correct enforcement and document the governing rule for evaluator reports.

Additional applications include rules coordinators testing how proposed rule changes alter enforcement outcomes, coaches and broadcast analysts verifying enforcement explanations, and expansion to other sports through The Call AI platform, which envisions parallel officiating assistants for soccer, basketball, baseball, hockey, and other sports. In each case, the innovation provides value through consistent, citable rulings that reduce ambiguity, shorten decision time, and accelerate official development.

SIP BRIEF PART 2

Evaluation Criteria

"The following questions will identify the successful completion of the project."

Create a list of **"Yes/No"** questions that will form the criteria for YOU to determine the success/completion of your project in terms of usability or usefulness of the product you will eventually prototype. (This list will NOT be used by your SME – it will act as your To-Do list).

Provide as many criteria as required to assure the quality of the project; each one needs to be expressed in such a way that it is measurable or observable and should be in the form of a **yes or no** question. Also, do NOT answer the questions. This is simply a list of what the criteria are.

Goals and Tasks Associated with the Project

List and describe the High-Level objectives of this project. These should include the targeted areas of completion that will guarantee the success of your project. (Each objective will eventually have several tasks associated with it.)

After giving the objectives, list and describe all tasks that will allow you to complete each objective. Tasks are the specific actions required to achieve the objectives. The Evaluation Criteria mentioned above will ensure that the quality is ultimately measurable or observable. Each task shall be described in detail and include transparency on the decision-making process. Use headings for each task to maintain clarity in the documentation. Use dates (if known) to specify when the tasks will be completed and implemented.

Note: This section must be revised prior to SIP408 to describe tasks as they actually occurred.

Description of Design Prototype

Describe the **design prototype** implementation. You should describe the platform on which the system will be built and provide directions on how to run the prototype (if necessary). Elaborate on the functionality of the parts that define your project. Effective descriptions will give the reader an understanding of what the design prototype will be, and how it relates to the final project.

Note: This section will be revised prior to SIP408 to describe the design prototype in its final form.

SIP Brief Part 3 – to be filled out in SIP409

Evaluation Plan

Provide a complete, paragraph-style description of the plan used to evaluate your project. This section should be a **description of the full plan** for how you will go about answering the “Evaluation Criteria” questions. Do not simply repeat the questions!

Note: This section must be revised in SIP409 to describe the full evaluation plan as it was actually implemented.

Project Completion Assessment

Note: This section must be completed in SIP409

Provide an in-depth description of the completion assessment of your project. Describe how well the completed components function and highlight the innovative facets of your design. This is sometimes known as a “Post-Mortem” or “Lessons-Learned Report”. A good approach for this section is to answer the following 4 questions: “What went right? What went wrong? What was learned throughout the process? What would be done differently if you had to do it again?”

Appendices

Note: This section must be completed in SIP408.

Include as appendices any supporting material for this project, including charts, graphs, and other data; images associated with the project; or other documentation (e.g., a game design document or read-me file). Include any prior art that was used such as U.S. Patent Documents, Foreign Patent Documents, or other sources. Remember that this section should only be a list of additional files, not the actual data of the files!

Use the following format:

Appendix letter: description of item – file name

Example...

Appendix A: Game design document – myGameDoc.docx

Appendix B: 3D render of primary character – mainCharacter.jpg